

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

How to Use the Standard Normal (Z) Table

This table gives the **area under the standard normal curve to the LEFT of a given Z-score**. It is a pre-computed lookup table so you don't need to solve the Normal distribution's integral by hand.

How the Table is Structured

- The left-most column gives the Z-score up to **one decimal place** (e.g. 1.3, -0.8, 2.0)
- The top row gives the **second decimal place** (.00 to .09)
- Where the row and column intersect is the area (probability) to the **left** of that exact Z-score

Z	.00	.01	.02	.03	.04
1.3	.90320	.90490	.90658	.90824	.90988

Example 1 — Area to the LEFT of Z = 1.33

Context: Heights of adult males follow $X \sim N(68, 3)$. What is $P(X > 72)$?

1. Convert to a Z-score: $Z = (72 - 68) / 3 = 1.33$
2. Split $Z = 1.33$ into row **1.3** and column **.03**
3. Find row 1.3, move across to column .03
4. Read off the value: **0.90824**
5. Interpretation: 90.824% of the area lies to the LEFT of $Z = 1.33$, so $P(Z \leq 1.33) = 0.90824$
6. Since we want the area to the RIGHT (taller than 72"): $1 - 0.90824 = \mathbf{0.09176}$ ($\approx \mathbf{9.18\%}$)

Example 2 — Area to the LEFT of Z = -0.85

1. Split $Z = -0.85$ into row **-0.8** and column **.05**
2. Find row -0.8 (upper half of table, negative Z-scores) and column .05
3. Read off the value: **0.19766**
4. Interpretation: Only 19.766% of the area lies to the left of $Z = -0.85$ — makes sense, since a negative Z-score means we're below the mean, so less than 50% of the curve is to the left.

Example 3 — Area BETWEEN Two Z-scores

Find the area between $Z = -0.85$ and $Z = 1.33$

1. Look up each Z-score individually: $\text{Area}(Z=1.33) = 0.90824$, $\text{Area}(Z=-0.85) = 0.19766$
2. Subtract the smaller from the larger, since both are measured from the left:
 $P(-0.85 \leq Z \leq 1.33) = 0.90824 - 0.19766 = \mathbf{0.71058}$ ($\approx \mathbf{71.06\%}$)
3. Interpretation: About 71% of the data lies between these two Z-scores.

Quick Reference — What Each Area Question Needs

You want...	Formula using table values
Area to the LEFT of Z	Read directly from the table
Area to the RIGHT of Z	$1 - (\text{table value})$
Area BETWEEN two Z-scores ($Z1 < Z2$)	$(\text{table value for } Z2) - (\text{table value for } Z1)$